ECEN4293 Homework 2

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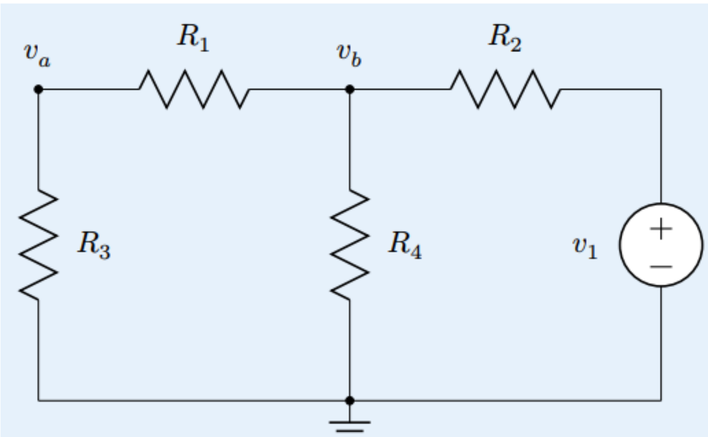
Problem 1: A Simple Circuit

Consider the circuit below.



Component	Value
v_1	10 V
R_1	1 kΩ
R_2	1 kΩ
R_3	9 kΩ
R_4	10 kΩ

(a) Using Kirchhoff's Current Law and Ohm's Law, write a linear system of equations in terms of v_a and v_b for this circuit.

$$\begin{bmatrix} \frac{-1}{R_1} & (\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_1}) \\ (\frac{-1}{R_1} - \frac{1}{R_3}) & \frac{1}{R_1} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} \frac{V_1}{R_2} \\ 0 \end{bmatrix}$$

(b) Write Python code to find v_a and v_b using `numpy.linalg.solve`. Clearly indicate the resulting node voltages in your answer.

$$V_a = 7.50V$$
$$V_b = 8.33V$$

(c) Write a Python function to implement Gaussian elimination with partial pivoting. Use this function to compute the answer. Does it match the node voltages found using `numpy.linalg.solve`?

Yes, the answers provided by the Gaussian elimination approach matched the answers provided by numpy.linalg.solve approach.

Problem 2: Heat Modeling

Suppose a square sheet of flat material is heated by its edges. The Poisson equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

describes how heat transfers through the material over time. This is a partial differential equation, but we can approximate the steady-state solution using linear algebra.

Suppose we divide the square material into a mesh of $m \times m$ evenly spaced points x_i, y_j.

The solution at each point can be approximated using a 5-point stencil:

$$\frac{1}{h^2}\left(u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{ij}\right) = 0$$

Writing an equation for each point results in a block linear system:

$$\frac{1}{h^2}\begin{bmatrix} T & I & & \\ I & T & I & \\ & \ddots & \ddots & \ddots \\ & & I & T \end{bmatrix}\begin{bmatrix} u_{1j} \\ u_{2j} \\ \vdots \\ u_{mj} \end{bmatrix} = \mathbf{b}$$

where I is an $m \times m$ identity matrix and T is the tridiagonal matrix

$$T = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -4 \end{bmatrix}$$

and $\mathbf{b}$ contains the boundary values.

Suppose the sheet has its center at the origin and corners at $(-1,-1)$ and $(1,1)$. The boundary heating function is

$$f(x,y) = 25 - x^2 + y^2$$

(a) Write a Python function to construct the $m \times m$ tridiagonal matrix T given m.

(b) Write a Python function that uses your tridiagonal matrix function and `np.identity` to construct the full matrix A.

(c) Write a Python function to compute the vector $\mathbf{b}$ using $f(x,y)$ for the boundary conditions.

(d) Write a Python function to compute the Cholesky factorization of a square symmetric matrix.

(e) For $m = 10$ and $m = 25$:

- Construct the linear system.

  For m =10

$$\begin{pmatrix} -121 & 30.25 & & & & \\ 30.25 & -121 & 30.25 & & & \\ & 30.25 & -121 & 30.25 & & \\ & & 30.25 & -121 & \ddots & \\ & & & \ddots & \ddots & 30.25 \\ & & & & 30.25 & -121 \end{pmatrix}$$

For m = 25

$$\begin{pmatrix} -676 & 169 & & & & \\ 169 & -676 & 169 & & & \\ & 169 & -676 & 169 & & \\ & & 169 & -676 & \ddots & \\ & & & \ddots & \ddots & 169 \\ & & & & 169 & -676 \end{pmatrix}$$

- Solve it using Cholesky factorization.

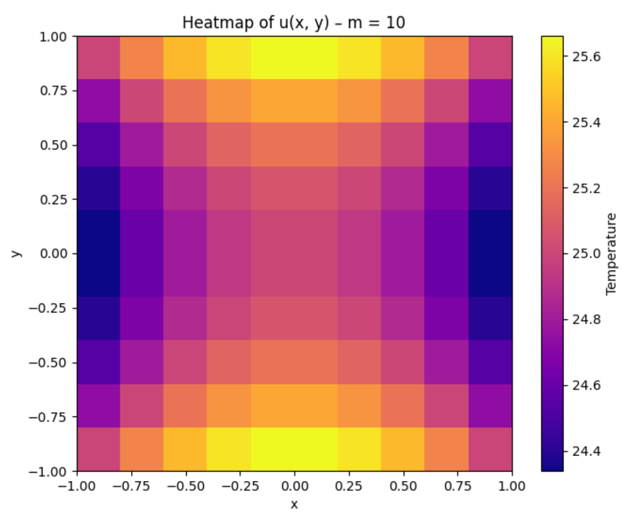
  For m = 10

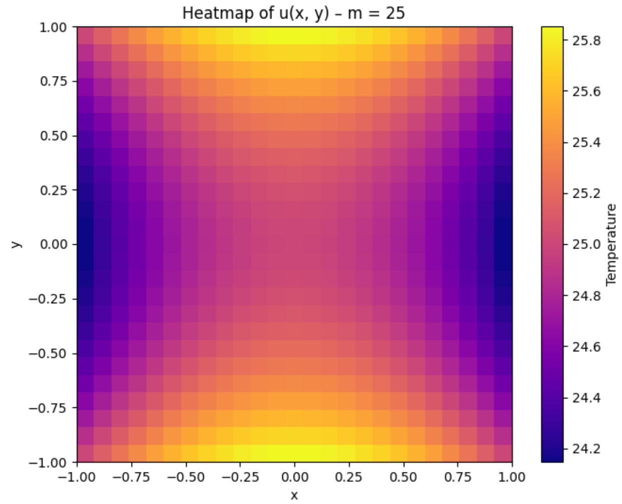
$$L = \begin{pmatrix} 11 & 0 & & & & & \\ -2.75 & 10.65 & 0 & & & & \\ -0. & -2.84 & 10.63 & 0 & & & \\ & & \ddots & \ddots & & & \\ & & & \ddots & 9.89 & & \\ & & & & -3.51 & 9.91 & \\ & & & & -0.16 & -3.43 & 10 \end{pmatrix}$$

  For m = 25

$$L = \begin{pmatrix} 26 & 0 & & & & & \\ -6.5 & 25.17 & 0 & & & & \\ -0. & -6.71 & 25.12 & 0 & & & \\ & & \ddots & \ddots & & & \\ & & & \ddots & 23.36 & & \\ & & & & -8.31 & 23.42 & \\ & & & & -0.37 & -8.11 & 23.64 \end{pmatrix}$$

- Plot a heatmap of the solution $u(x, y)$.




Heatmap of u(x, y) – m = 25
y
x
Temperature
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
25.8
25.6
25.4
25.2
25.0
24.8
24.6
24.4
24.2